

Prediction of Student Academic Performance Using Machine Learning

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Abstract—Predicting student performance is a significant machine learning application within educational data mining that allows institutions to predict performance and make data-informed decisions. This thesis details a Machine Learning-Based Student Performance Prediction System for predicting pass/fail status of a student and final course grade of students based on student's related academic and behavioral features. The presented system covers a full machine learning pipeline consisting of data pre-processing, feature transformation, model training, evaluation and prediction. Various classification and regression models were used and evaluated to determine appropriate prediction models for classification and regression tasks, respectively. Furthermore, explainable artificial intelligence (XAI) features (feature importance and SHAP value) were used for better model interpretability. From our experiment results, the best performance for classification was achieved by Random Forest classifier in predicting the pass/fail of a student with 79.17 percent accuracy and F1-score of 0.8824, whereas for predicting grade, the lowest error for prediction was obtained by Linear Regression with a RMSE of 9.60. This shows that machine learning methods could successfully predict student classification while there is a room for improvement for student grade prediction.

Index Terms—Machine Learning, Educational Data Mining, Student Performance Prediction, Classification, Regression, Explainable Artificial Intelligence (XAI)

I. INTRODUCTION

Availability of many educational data and improvement in machine learning techniques has paved the path to improve decisions for educational process and students support system. Students related information such as attendance, study habits, achievement, engagement in the learning process is collected by educational institutions in regular basis. By analyzing such data meaningful information is provided to educational institutions for decision making instead of making decision based on intuition [1]. Therefore Educational Data Mining (EDM) and Learning Analytics have become very popular research fields dealing with the extraction of useful information from educational data for the improvement of teaching and learning processes [2].

Student academic performance prediction is one of the important applications of machine learning in education. Identifying potential struggling students helps educators make timely interventions, individualized coaching and more support for their learning. Predicting systems can help students achieve more success in their academics, thus enhancing their academic planning.

While educational data is more widely accessible than ever before, accurately predicting students' academic achievement is no simple task. Student performance is dependent on several interconnected variables, such as attendance, study time, past academic records, parental involvement, and participation in extracurricular activities. Existing forms of assessment focus on previous performance, and seldom have the functionality to predict future performance. Thus, there is a demand for intelligent, data-driven systems that can analyze students' performance predictors and effectively predict student achievement [1] [3].

The purpose of this project is to build a ML-Based Student Performance Prediction System where the classification can predict pass or fail and the regression can predict the final grade of the student. Various ML algorithms will be utilized and trained and evaluated. Techniques on XAI, such as explanation with SHAP and feature importance will be used as an enhancement to model interpretability [4].

The project's scope includes Data pre-processing, feature engineering, model training and development, model evaluation, prediction, explainability. The scope is limited to the data and machine learning models used in the suggested methodology. Integrating the machine learning approach, drawing from predictive modeling and explainable AI, this project demonstrates the contribution of machine learning in learning analytics to enable stakeholder understanding of factors impacting students' academic success.

II. RELATED WORK

Prediction of student performance is a trending topic in both Educational Data Mining (EDM) and Learning Analytics. Due

to increased availability of educational data, it has become possible for researchers to make use of machine learning and data mining methods for student behavior, performance prediction and decision making. Romero and Ventura [2] surveyed the field of EDM and noted an increase in the use of predictive models for educational applications and explained that such techniques make it possible to uncover learning patterns.

Cortez and Silva [5] made a major contribution by using data mining to predict the performance of secondary school students on their academic subjects, taking into account some demographic, social and academic attributes. They demonstrated that students' previous academic success, study habits, and family-related features are the major factors responsible for good performance. To this day, this research is still one of the most quoted in the field of academic performance prediction.

Another research by Baker and Inventado [3] underscored the role of EDM and LA in revealing students who are prone to failure. In their study, they mentioned that predictive models will contribute to a data-based educational system that enables early intervention with students performing poor.

In [6], a detailed review of machine learning methods that can be used in predicting students' performance such as decision trees, random forests, artificial neural networks and support vector machines was undertaken. Shahiri et al. [6] claimed that machine learning algorithms are indeed a useful way in predicting students' academic performance and that these methods could be very useful for institutions in education planning and student support.

Random Forest is a prominent machine learning algorithm gaining popularity owing to its robustness and ability to deal with intricate datasets [7] and predict well on the data. Machine learning algorithms such as decision tree, logistic regression are widely utilized in prediction with regard to the academic context. These algorithms have proven to work with classification problems that include student passing, failing or students that would need help in the academic aspect.

Beyond predictive accuracy, interpretability of the machine learning models is another aspect of critical importance in the educational applications. Users in the educational context need explanation for the prediction derived from a model before using them in decision-making. SHAP (SHapley Additive exPlanations) proposed by Lundberg and Lee [4] has successfully provided a unified framework for interpreting machine learning predictions by generating feature-level explanations.

With these research articles in mind, the present work focuses on developing a "Machine Learning-Based Student Performance Prediction System". In this project, both classification and regression techniques are utilized and experimented, and various machine learning algorithms are deployed and evaluated for the task of student's pass/fail status and the final grade in courses. Moreover, SHAP-based explanation techniques are incorporated in this system for the interpretability purpose, where key features that contribute to student's academic performance can be investigated.

III. DATASET AND PREPROCESSING

This project used a tabular student performance dataset stored in a CSV file named `student_performance_updated_1000.csv`.

The dataset was loaded from Google Drive and initially contained 1000 student records with 12 attributes. These attributes included student identity information, demographic information, attendance, study habits, previous academic performance, parental support, online class participation, and final grade. A summary of the dataset is presented in Table I.

The main objective of this project was to perform two prediction tasks. First, a classification task was performed to predict whether a student would pass or fail. Second, a regression task was performed to predict the student's final grade. The original `FinalGrade` column was used to create the classification target. Students with `FinalGrade` greater than or equal to 70 were labeled as Pass, while students with grades below 70 were labeled as Fail. The distribution of Pass and Fail classes after target creation is shown in Fig. 2.

During dataset inspection, missing values were found in several columns. Since the `FinalGrade` column was required for both target creation and regression prediction, rows with missing `FinalGrade` values were removed. After this cleaning step, the final working dataset contained 960 records. Duplicate rows were also checked, and no duplicate records were found.

For model training, non-predictive and leakage-related columns such as `StudentID`, `Name`, and `FinalGrade` were removed from the input feature set. The remaining features were used as model inputs. The final feature set contained 9 input features, including 6 numerical features and 3 categorical features, as listed in Table II. Numerical missing values were handled using mean imputation, while categorical missing values were handled using most frequent value imputation. Numerical features were standardized using `StandardScaler`, and categorical features were converted into numerical form using `OneHotEncoder`. The overall preprocessing workflow is illustrated in Fig. 1.

The dataset was divided into training and testing sets using an 80:20 train-test split. Stratified splitting was applied for the classification target so that the Pass and Fail class distribution remained similar in both training and testing sets. After splitting, the training set contained 768 records and the testing set contained 192 records. No data augmentation technique was applied because the dataset was tabular, not image or text based.

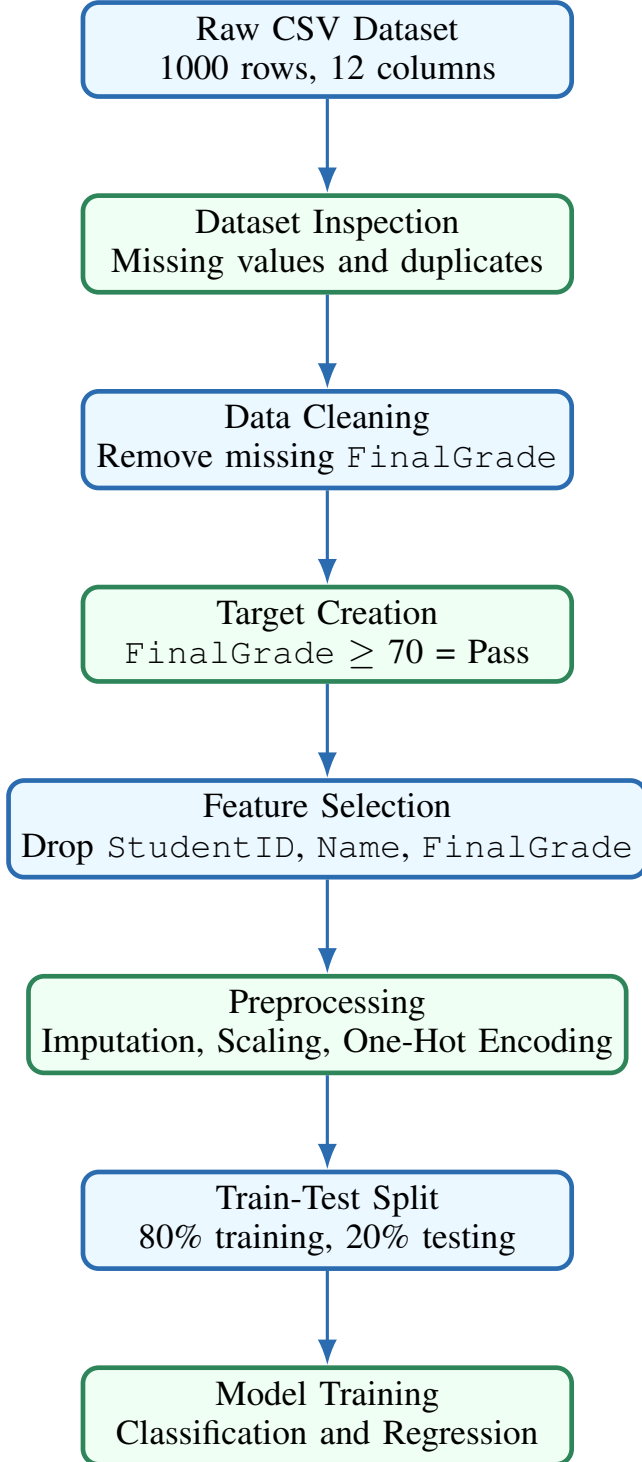


Fig. 1. Dataset preprocessing workflow

TABLE I
DATASET SUMMARY

Property	Description
Dataset type	Tabular CSV dataset
File name	student_performance_updated_1000.csv
Initial dataset size	1000 rows and 12 columns
Rows after cleaning	960 rows
Input features	9
Numerical features	6
Categorical features	3
Classification target	Performance Pass/Fail
Regression target	FinalGrade
Train-test split	80% training and 20% testing
Training records	768
Testing records	192

TABLE II
FINAL FEATURES USED FOR MODEL TRAINING

Feature Type	Features
Numerical features	AttendanceRate, StudyHoursPerWeek, PreviousGrade, ExtracurricularActivities, Study Hours, Attendance (%)
Categorical features	Gender, ParentalSupport, Online Classes Taken
Removed columns	StudentID, Name, FinalGrade
Engineered target	Performance: Pass = 1, Fail = 0

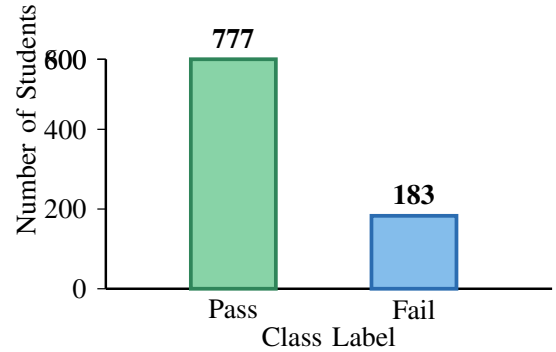


Fig. 2. Pass and Fail class distribution after target creation

IV. SYSTEM DESIGN AND METHODOLOGY

The proposed system was designed as a machine learning-based student performance prediction pipeline. The system takes student-related academic and behavioral features as input and performs two prediction tasks. The first task is a classification task that predicts whether a student will Pass or Fail. The second task is a regression task that predicts the student's final grade. The overall system architecture is shown in Fig. 3.

This project uses structured student records. That's why, machine learning algorithms such as Logistic Regression, Decision Tree, Random Forest, and Linear Regression are more suitable and computationally efficient.

For the classification task, three models were trained and evaluated: Logistic Regression, Decision Tree Classifier, and Random Forest Classifier. Logistic Regression was used as a simple baseline model because it is easy to interpret and works well for binary classification problems. Decision Tree was selected because it can capture non-linear relationships and provide interpretable decision rules. Random Forest was selected because it combines multiple decision trees and usually provides better generalization by reducing overfitting.

For the regression task, three models were trained and evaluated: Linear Regression, Decision Tree Regressor, and Random Forest Regressor. Linear Regression was used as a baseline model for predicting continuous final grades. Decision Tree Regressor was used to capture non-linear patterns in the data, while Random Forest Regressor was used as an ensemble method to improve prediction stability.

The implementation was carried out using Python in Google Colab. The main libraries used in this project were pandas and numpy for data handling, matplotlib for visualization, scikit-learn for preprocessing, model training, and evaluation, SHAP for explainable AI analysis, and joblib for saving trained models. A summary of the tools and frameworks is given in Table III.

The training procedure followed a pipeline-based approach. First, the dataset was divided into training and testing sets using an 80:20 train-test split. For classification, stratified splitting was used to preserve the Pass and Fail class ratio. Then, numerical and categorical preprocessing steps were applied using a ColumnTransformer. Numerical features were imputed using mean imputation and scaled using StandardScaler. Categorical features were imputed using the most frequent value and encoded using OneHotEncoder. After preprocessing, each machine learning model was trained using the training set and evaluated using the testing set.

For classification evaluation, accuracy, precision, recall, F1-score, fail recall, pass recall, and 5-fold stratified cross-validation F1-score were used. For regression evaluation, mean absolute error (MAE), root mean squared error (RMSE), R-squared score, and 5-fold cross-validation R-squared score were used. The main algorithms and their hyperparameters are listed in Table IV.

TABLE III
TOOLS AND FRAMEWORKS USED

Tool/Framework	Purpose
Python	Main programming language
Google Colab	Development and execution environment
pandas	Dataset loading and data manipulation
numpy	Numerical operations
matplotlib	Data visualization
scikit-learn	Preprocessing, model training, and evaluation
SHAP	Explainable AI and feature impact analysis
joblib	Saving trained machine learning models

TABLE IV
MODELS AND HYPERPARAMETERS

Model	Main Hyperparameters
Logistic Regression	max_iter=1000, class_weight=balanced, random_state=42
Decision Tree Classifier	max_depth=6, min_samples_split=10, min_samples_leaf=5, class_weight=balanced, random_state=42
Random Forest Classifier	n_estimators=300, max_depth=8, min_samples_split=10, min_samples_leaf=4, class_weight=balanced_subsample, random_state=42
Linear Regression	Default parameters
Decision Tree Regressor	max_depth=8, min_samples_leaf=5, random_state=42
Random Forest Regressor	n_estimators=200, max_depth=10, min_samples_leaf=3, random_state=42

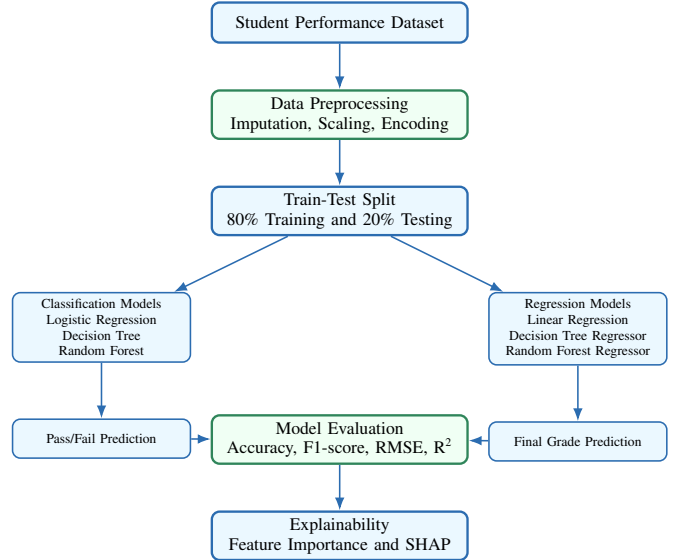


Fig. 3. System architecture of the student performance prediction model

V. IMPLEMENTATION AND RESULTS

The proposed student performance prediction system was implemented using Python in Google Colab. The complete implementation process included dataset loading, data cleaning, preprocessing, feature transformation, train-test splitting, model training, model evaluation, explainability analysis, and final prediction generation. The dataset was loaded from Google Drive and processed using Python libraries such as pandas, numpy, matplotlib, scikit-learn, and SHAP. The trained models were saved using joblib so that they could be reused for future prediction.

The experimental setup followed an 80:20 train-test split. In this setup, 80% of the cleaned data was used for training

and 20% was used for testing. For the classification task, stratified splitting was applied so that the Pass and Fail class distribution remained similar in both training and testing sets. The system performed two major tasks. The first task was Pass/Fail classification, and the second task was final marks prediction using regression models.

For Pass/Fail classification, three models were trained and evaluated: Logistic Regression, Decision Tree Classifier, and Random Forest Classifier. These models were evaluated using accuracy, precision, recall, F1-score, Fail recall, Pass recall, and cross-validation F1-score. The overall comparison of the classification models is shown in Fig. 4. From the result, Random Forest achieved the best classification performance with an accuracy of 79.17% and an F1-score of 0.8824. It also achieved very high Pass recall, which means that the model correctly identified most students who passed.

The confusion matrix of the Random Forest classifier is shown in Fig. 5. The model correctly predicted 150 students as Pass and 2 students as Fail. However, 35 actual Fail students were predicted as Pass, and 5 actual Pass students were predicted as Fail. This result shows that the model performed well for the majority Pass class but struggled to identify the minority Fail class. The main reason for this limitation is class imbalance, as the dataset contained more Pass cases than Fail cases.

For final marks prediction, three regression models were trained and evaluated: Linear Regression, Random Forest Regressor, and Decision Tree Regressor. The regression models were evaluated using Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). The marks prediction error comparison is shown in Fig. 7. Linear Regression achieved the lowest RMSE value of 9.60, making it the best regression model for this experiment. Random Forest Regressor produced a very close error value, while Decision Tree Regressor showed comparatively higher error.

The actual versus predicted marks plot for Linear Regression is shown in Fig. 8. The dashed diagonal line represents perfect prediction, where the actual and predicted marks are equal. The predicted values are mostly clustered within a limited range. This indicates that the model can estimate the general academic performance of students, but it cannot perfectly predict every individual mark. This also explains why the regression task has room for improvement.

To interpret the classification model, feature importance analysis was performed using the Random Forest classifier. The feature importance plot is presented in Fig. 9. The result shows that attendance percentage, study hours, study hours per week, previous grade, and attendance rate were among the most influential features. This indicates that student engagement, regular attendance, study time, and previous academic performance strongly influenced the prediction of student success.

In addition to feature importance, SHAP analysis was used to improve model interpretability. The SHAP summary plot is shown in Fig. 10. SHAP values explain how each feature contributes positively or negatively to the model output. Fea-

tures such as study hours per week, attendance percentage, study hours, attendance rate, and previous grade had noticeable effects on the prediction. This explainability step makes the system more useful because it helps users understand why a model makes a specific prediction.

After selecting the best-performing models, the system was tested using a sample student record. The sample output is shown in Fig. 11. The system displayed the actual status, actual marks, predicted status, predicted marks, Fail probability, and Pass probability. In the sample case, the actual status was Pass and the system also predicted the status as Pass. The predicted marks were close to the actual marks, showing that the system can generate both classification and regression outputs for an individual student.

The system also generated actionable improvement suggestions based on the student's current feature values. The recommendations included improving attendance rate, maintaining attendance above 85%, increasing daily study hours, improving academic fundamentals, and participating in online classes regularly. This makes the system more useful because it not only predicts student performance but also provides guidance for possible improvement.

Based on the experimental results, the final model selection process is summarized in Fig. 6. Random Forest was selected as the best classifier because it achieved the highest classification accuracy and F1-score. Linear Regression was selected as the best regressor because it achieved the lowest RMSE value among the regression models.

Overall, the experimental results show that Random Forest was the most effective model for the Pass/Fail classification task, while Linear Regression performed best for final marks prediction. The results also indicate that attendance, study time, and previous academic performance were important factors affecting student performance. However, the confusion matrix shows that the model had difficulty identifying Fail cases because of class imbalance. Therefore, future improvements may include collecting a larger and more balanced dataset, applying class balancing techniques, and adding more student-related features to improve the detection of at-risk students.

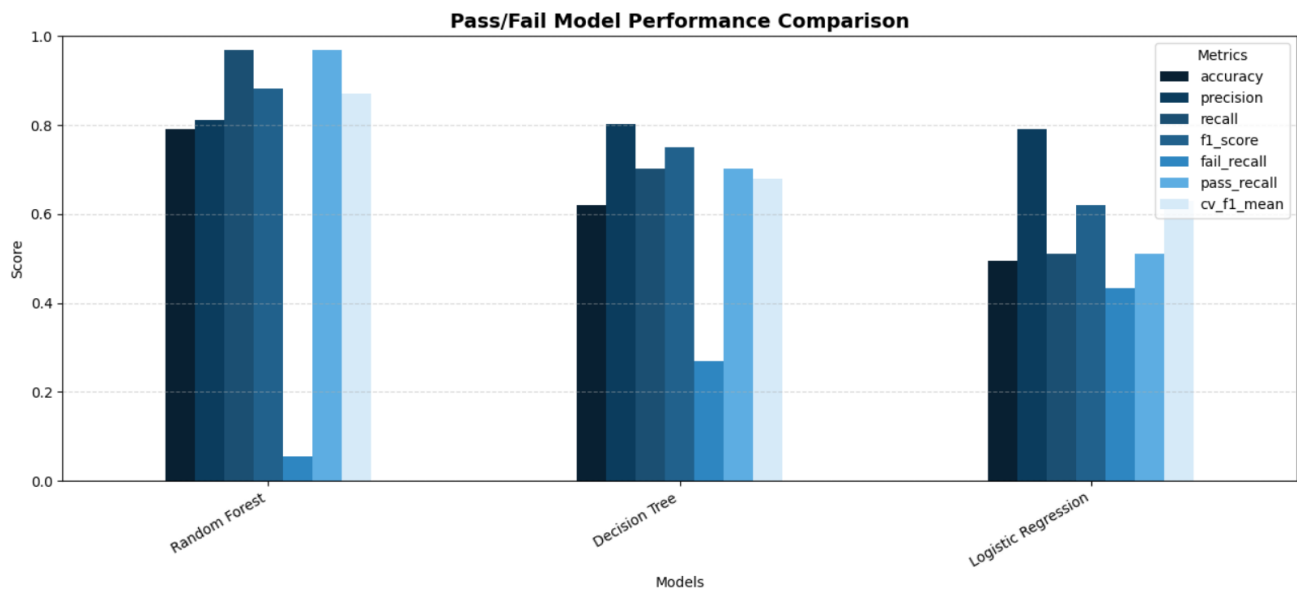


Fig. 4. Pass/Fail model performance comparison using multiple evaluation metrics

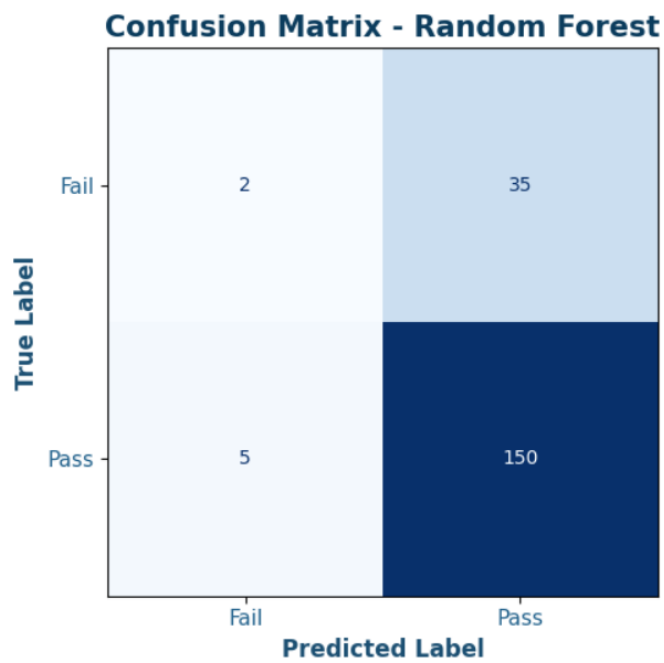


Fig. 5. Confusion matrix of the Random Forest classifier

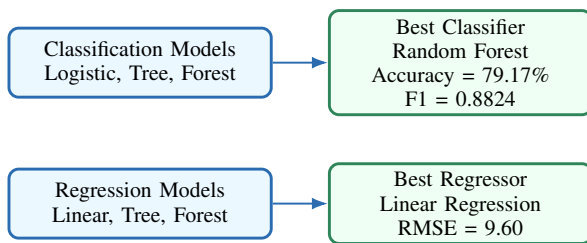


Fig. 6. Final model selection based on experimental results

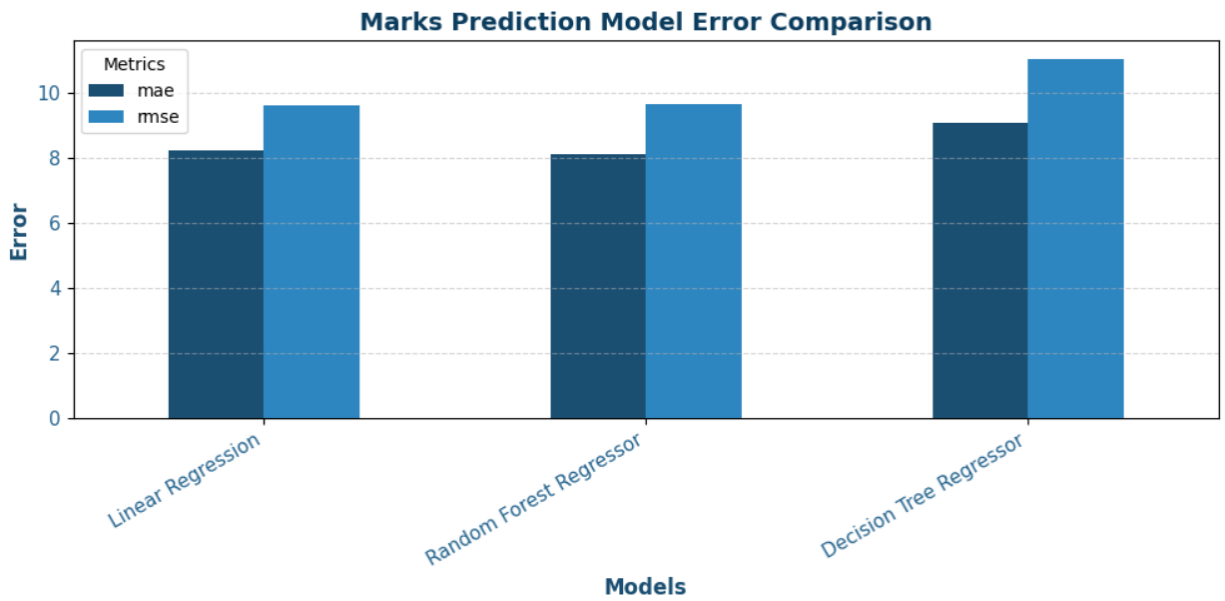


Fig. 7. Marks prediction model error comparison using MAE and RMSE

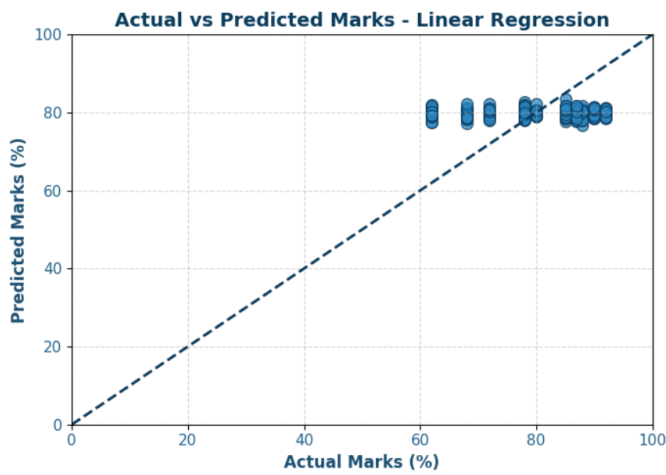


Fig. 8. Actual versus predicted marks for Linear Regression



Fig. 10. SHAP summary plot showing feature impact on model prediction

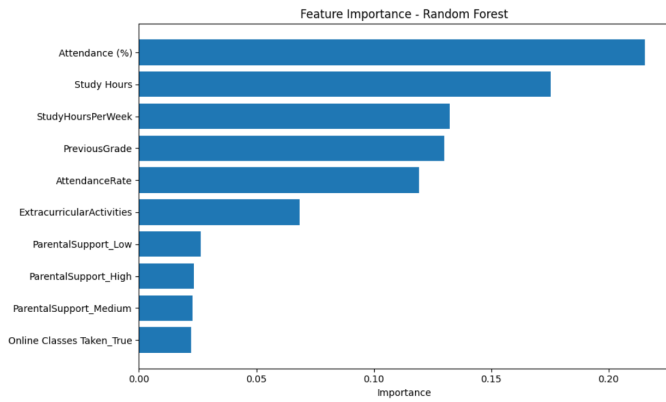


Fig. 9. Feature importance analysis of the Random Forest classifier

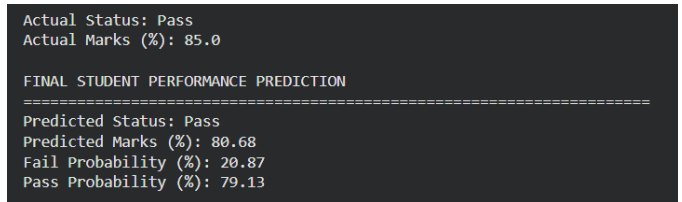


Fig. 11. Sample final student performance prediction output

VI. DISCUSSION

Three classification models were used in this project: Logistic Regression, Decision Tree, and Random Forest. After evaluating their performance, Random Forest was found to be the best model. It achieved an accuracy of 79.17 percent and an F1-score of 88.24 percent, which were higher than the scores of the other models. The model also obtained the highest cross-validation F1-score (0.8703), indicating that its performance was stable and consistent across different data splits.

Random Forest performed well because it combines the predictions of multiple decision trees instead of relying on a single tree. This approach helps reduce overfitting and allows the model to capture more complex patterns in the dataset. As a result, it was able to make more accurate predictions than Logistic Regression and Decision Tree. Logistic Regression showed the weakest performance because it assumes a linear relationship between the features and the target variable, while student performance is often influenced by multiple complex factors.

The results also show that the best model achieved a very high Pass Recall of 96.77 percent. This means that the model was able to correctly identify most students who passed. Therefore, it can be considered reliable for recognizing successful students. However, the Fail Recall was only 5.41 percent, which indicates that the model struggled to identify students who failed. Many failing students were incorrectly predicted as passing students.

One of the main challenges in this project is the imbalance in the dataset. The number of students who passed was much higher than the number of students who failed. Because of this imbalance, the model learned the patterns of the majority class more effectively and had difficulty learning the characteristics of the minority class. This explains why the pass recall was very high while the fail recall remained low.

Another limitation is that the dataset may not contain all the factors that affect student performance. Variables such as family background, learning environment, motivation level, classroom participation, and personal circumstances were not included in the model. The absence of these factors may have reduced the model's ability to make more accurate predictions.

Several improvements can be made in future work. A larger and more balanced dataset could help the model learn both pass and fail cases more effectively. Additional student-related features could also be included to improve prediction quality. Furthermore, techniques such as data balancing, feature engineering, and advanced machine learning algorithms could be applied to achieve better overall performance and improve the detection of at-risk students.

VII. CONCLUSION

This project successfully developed a machine learning-based system for predicting student pass/fail outcomes. Three classification models—Logistic Regression, Decision Tree, and Random Forest—were trained and evaluated using different performance metrics. Among them, Random Forest

achieved the best overall performance with an accuracy of 79.17 percent and an F1-score of 88.24 percent, making it the most effective model for this dataset.

The results indicate that machine learning techniques can be useful for analyzing student performance and identifying patterns that influence academic success. Although the model showed strong performance in predicting students who passed, it was less effective in identifying students who failed, highlighting the need for further improvement in handling class imbalance and feature selection.

Overall, the project demonstrates the potential of data-driven approaches in the education sector. With additional data, more relevant features, and advanced machine learning techniques, the prediction accuracy can be further improved. In the future, this system could be used by schools, colleges, and universities to monitor student performance, identify at-risk students at an early stage, and provide timely academic support. Such applications can help educational institutions make better decisions and improve overall student outcomes.

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